Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A wall is 50 m away. An echo returns after 0.5 s. A student divides one-way distance by this time. Find the correct speed and explain the student's error.

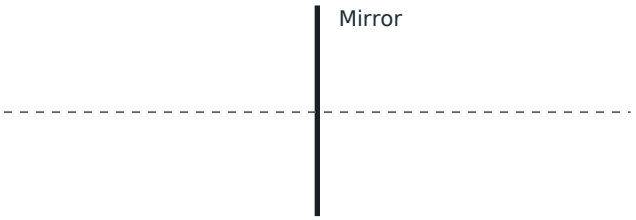
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; measure angles from the normal. The image is virtual.

Worked reasoning: An object 2 m in front of an ideal plane mirror has a virtual image 2 m behind it. The object-image separation is 4 m; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

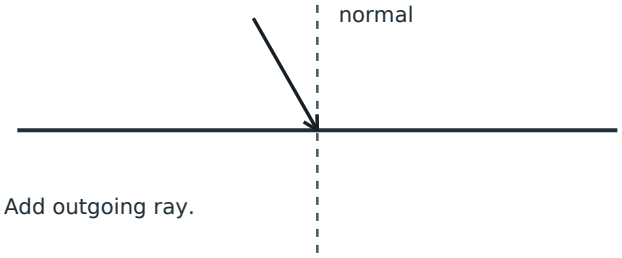


Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw the normal at the mirror. Complete a reflected ray for an incident ray 30° from normal.

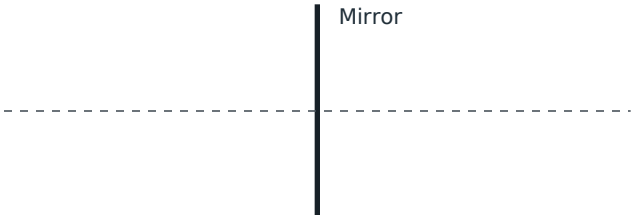


Add outgoing ray.

Marzano check - circle: ① ② ③

04 Apply the physics

A small lamp is 2 m in front of a plane mirror. Mark the lamp and its virtual image on the model. Find image distance behind the mirror and the full lamp-to-image separation.



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

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Continue down each column.

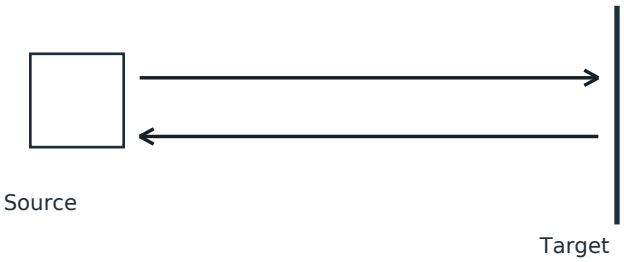
05 Explain your reasoning

Why is the mirror image called virtual rather than a real object behind the glass?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

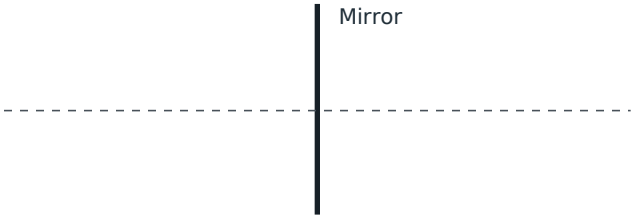
A reflected signal returns after 4 s. Name the extra quantity needed to calculate distance and explain why this is a round trip.
Label the outgoing and returning paths.



Marzano check - circle: 1 2 3

07 Exit check

Incidence angle 45° from normal. Find reflection angle.

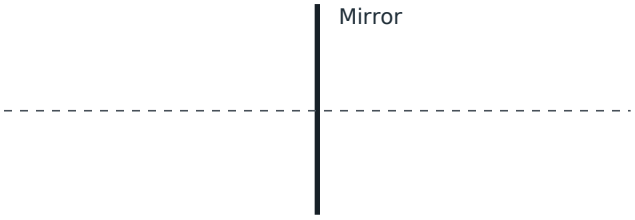


Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Move the object 0.5 m closer to the mirror. Find the new image distance and object-image separation.



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3